



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI SIVASAKTHI AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

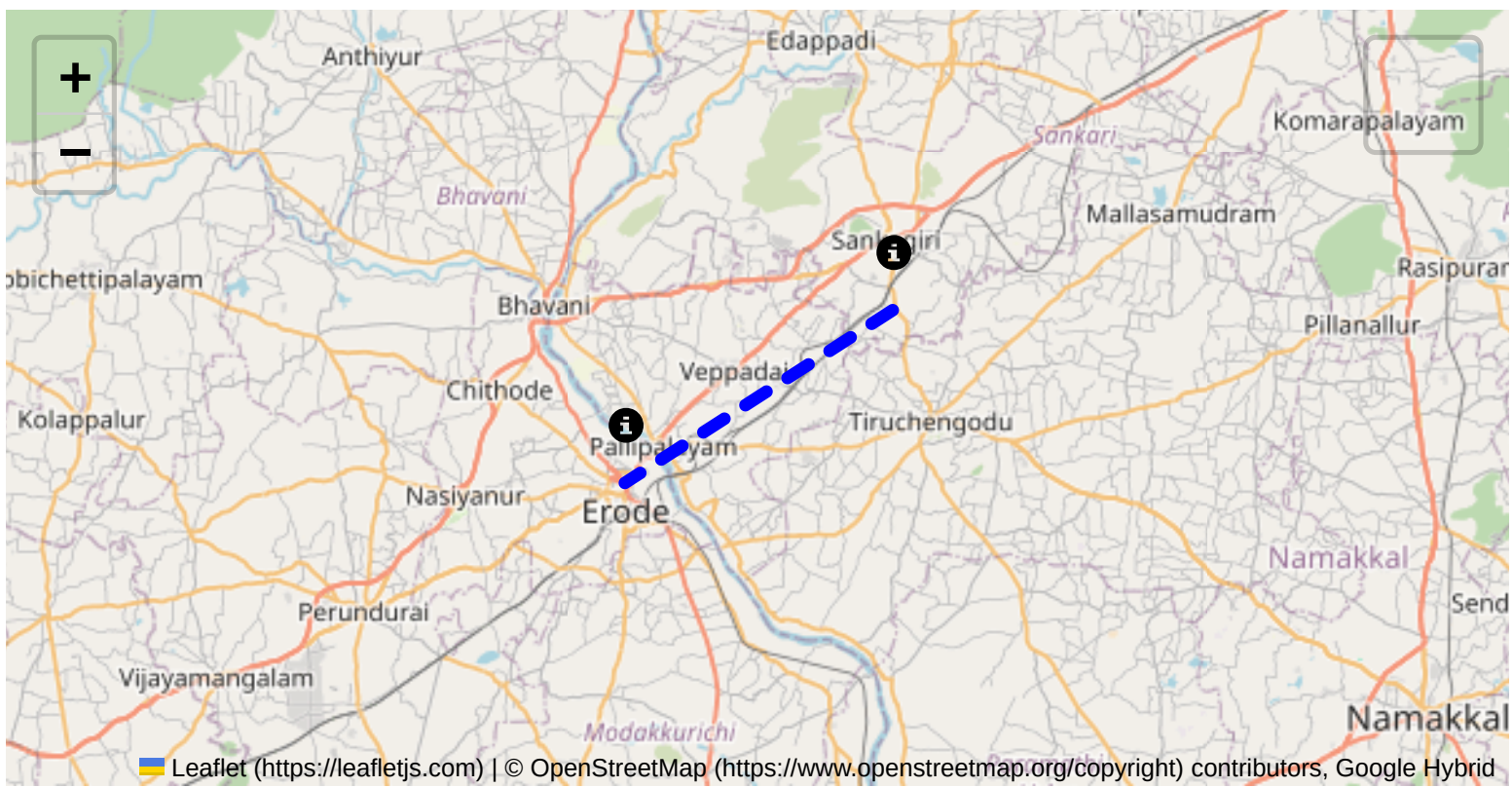
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

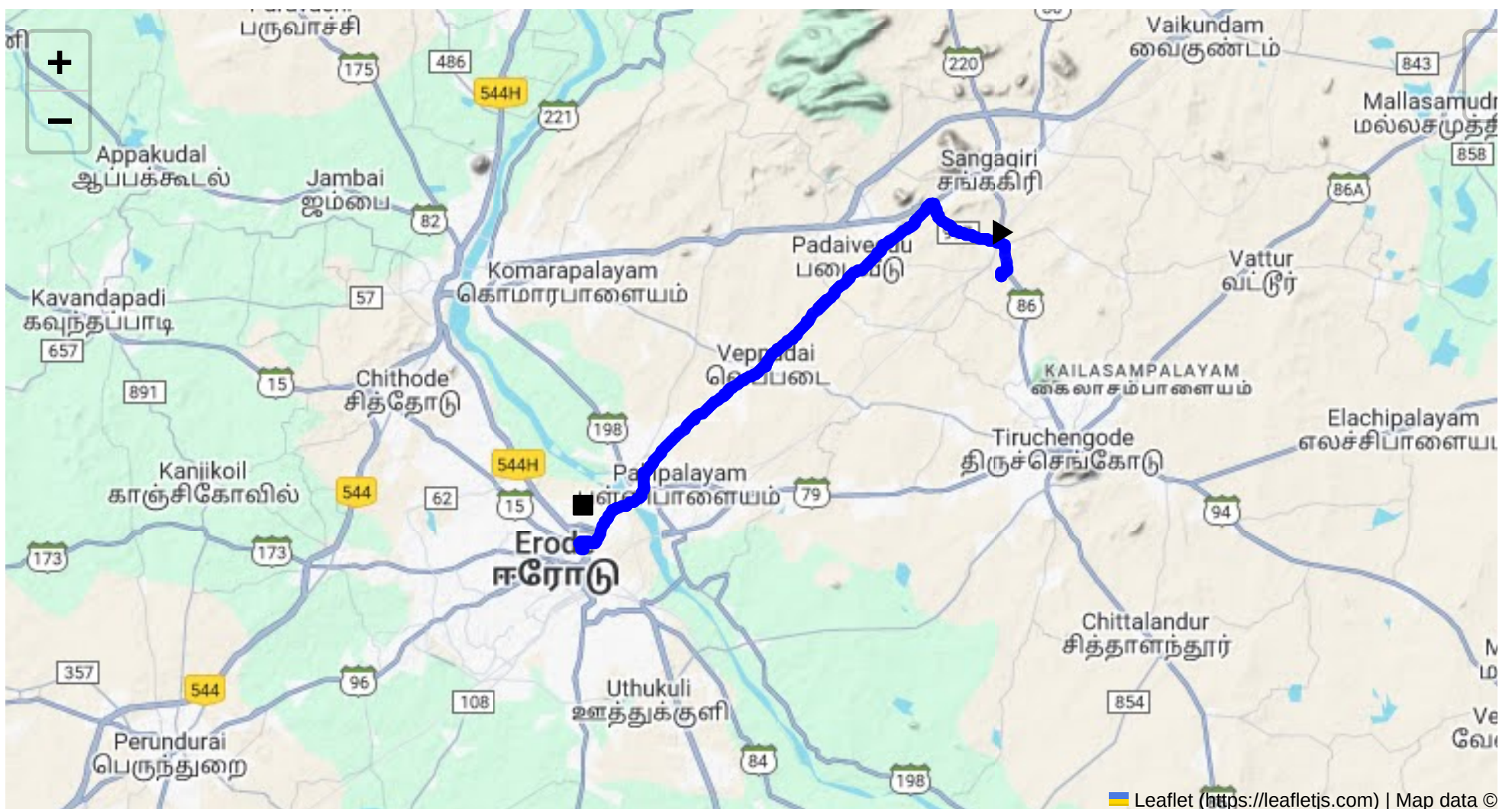
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 24.49 km
Estimated Duration: 0.6 hours
Adjusted Duration (Heavy Vehicle): 0.8 hours
Start: (11.4381, 77.8734)
End: (11.344964, 77.726338)



Welcome to the Journey Risk Management Study

1. Route Overview: The route from CVQF+23W, Sangagiri to 484B, Sathy Rd, Erode is approximately 24.49 kilometers, primarily traversing through local highways and smaller roads. The journey goes predominantly via NH544, a key regional corridor, and involves transitioning onto smaller urban roads as you approach Erode.

2. Weather Conditions and Hazards: Tamil Nadu generally experiences a tropical climate. Seasonal weather includes:

- **Summer (March to June):** High temperatures, potentially leading to engine overheating. Dust storms can reduce visibility.
- **Monsoon (July to September):** Heavy rainfall can cause waterlogging and slippery road conditions. Ensure vehicles are equipped for such conditions.
- **Post-monsoon/Winter (October to February):** Pleasant weather with minimal hazards but fog can occur in the mornings.

3. Traffic Patterns:

- **Peak Hours:** Morning (8:00-10:00 AM) and Evening (5:00-7:00 PM) are peak traffic times around urban centers like Erode.
- **Congestion-Prone Areas:** Highway intersections and toll plazas can experience congestion, particularly near city limits and entrances.

4. Road Quality and Infrastructure:

- **Highways (NH544):** Generally well-maintained with clear signage but can have occasional maintenance work which might cause delays.
- **Urban Roads:** Smaller roads entering Erode can have potholes and less maintenance, especially post-monsoon.

5. Alternative Routes for Emergencies:

- **via Athur:** An alternative route bypassing some urban congestion is possible through state highways SH86 and connecting roads, though it might extend travel time.

6. Local Regulations for Hazardous Material Transport:

- Transport of hazardous materials in India requires adherence to the Motor Vehicles Act and obtaining necessary permits. Movements are often restricted to non-peak hours or need permissions for urban entry.

7. Historical Incidents:

- Past incidents suggest occasional lorry overturns during monsoon due to slippery conditions. There have been no significant hazardous material spills noted publicly in the immediate route area, but vigilance is recommended.

8. Environmental Considerations:

- The route passes through agricultural areas; hence, spills could affect crop lands. There are no noted wildlife sanctuaries en route, but the region supports many local water bodies.

9. Communication Coverage:

- Coverage is generally reliable along NH544, but rural segments and transition routes might experience brief communication gaps, particularly in less populated areas.

10. Emergency Response Times:

- **Highways:** Emergency services such as ambulances and fire tend to respond quicker (within 30 minutes).
- **Urban Areas:** Response time might improve to about 20 minutes due to proximity to services.
- **Rural Segments:** Might experience delays up to 45 minutes due to distance from emergency facilities.

11. Overall Summary of Risk Assessment:

- **Risk Level: Moderate:** While the route is generally straightforward, seasonal weather and road conditions present potential risks, particularly in monsoon.
- **Recommendations:** Ensure vehicle maintenance (brakes, tires, wipers) especially pre-monsoon. Monitor weather forecasts, and plan journeys to avoid peak hours and potential congestion points. Ensure all regulatory permits are in place and emergency contacts are readily accessible.

Driver training focusing on regional driving challenges and familiarization with alternative routes would enhance safety and efficiency.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.37086, 77.74828	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44898, 77.87410	30 KM/Hr
6	Turn	Medium	11.45348, 77.85706	30 KM/Hr
7	Turn	Medium	11.45352, 77.85698	30 KM/Hr
8	Turn	Medium	11.46318, 77.84913	30 KM/Hr
9	Turn	Medium	11.46312, 77.84886	30 KM/Hr
10	Turn	High	11.34678, 77.73104	15 KM/Hr
11	Blind Spot	Blind Spot	11.34680, 77.72584	10 KM/Hr
12	Blind Spot	Blind Spot	11.34492, 77.72706	10 KM/Hr
13	Turn	High	11.34508, 77.72653	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Cheran Hospital	11.3680302, 77.7481976	30 km/h	Medium
2	pharmacy	Pharmacy	11.3652161, 77.7486375	30 km/h	Medium
3	hospital	Sri Kumaran E N T Hospital	11.3657814, 77.7477763	30 km/h	Medium
4	hospital	Sri Krishna Hospital	11.3656648, 77.7477783	30 km/h	Medium
5	hospital	Visaalam Hospitals	11.3651528, 77.747808	30 km/h	Medium
6	police	Police Station	11.3630177, 77.7476505	30 km/h	Medium
7	hospital	Government Hospital	11.3608508, 77.747453	30 km/h	Medium
9	hospital	Sri Hari Sakthivel Hospital	11.3526901, 77.7323055	30 km/h	Medium
10	hospital	Senthil Prakash Hospital	11.352639, 77.732267	30 km/h	Medium
11	hospital	V K Hospital	11.3513792, 77.7313976	30 km/h	Medium
12	hospital	Dhanvantri Critical Care Center	11.3504266, 77.7311274	30 km/h	Medium
13	hospital	Dhanvantri Critical Care Centre	11.35099, 77.730508	30 km/h	Medium
14	hospital	Indrajith Yoga Nature Cure Hospital	11.3499973, 77.732974	30 km/h	Medium
16	hospital	Kumudha Hospital	11.3467172, 77.7306797	30 km/h	Medium
17	hospital	City Hospital	11.3451151, 77.7300887	30 km/h	Medium
18	clinic	K.B.N. Nursing Home	11.344835, 77.7310771	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	Ssri Valliappa Vidhayalayam mat hr school	11.4004878, 77.7794381	30 km/h	Medium
8	marketplace	Weekly Market	11.3605669, 77.7488199	30 km/h	Medium
15	school	Municipality Girls Higher secondary school	11.3495618, 77.7325011	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37086, 77.74828



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45348, 77.85706



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46312, 77.84886



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.34678, 77.73104



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.34680, 77.72584



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.34492, 77.72706



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.34508, 77.72653
